Problem Chosen	2026 MCM/ICM Summary Sheet	Team Control Number
B		2602787

Breaking the Bottleneck of Earth-Moon Transportation: A Multi-objective Framework for Sustainable Large-scale Lunar Colonization

Summary

To support the establishment and long-term development of a future lunar colony accommodating 100,000 inhabitants, this paper systematically compares the overall performance of different transport strategies in terms of construction efficiency, economic cost, and sustainability, focusing on large-scale material transport between Earth and the Moon. Three scenarios are examined: a Galaxy Port scheme relying solely on space elevator systems, a scheme utilising existing rocket bases exclusively, and a hybrid transport scheme combining both approaches.

To address the substantial material transport demands during lunar base construction, time and cost models are established under transport capacity constraints, and the conflicting objectives of minimising construction duration and total expenditure are integrated within a multi-objective mixed-integer linear programming framework. A normalised weighting approach generates feasible solution sets and corresponding Pareto frontiers, quantitatively illustrating time–cost trade-offs under varying transport proportions. Results indicate that, relative to single-mode transport, hybrid strategies significantly shorten the overall construction period while introducing only limited incremental costs, yielding superior performance among the three schemes.

To enhance applicability under real-world conditions, the robustness of the transport system is further examined under non-ideal operational scenarios, assessing the impacts of reduced efficiency and delays. In addition, a system dynamics approach is employed to model annual water demand for long-term colony operation, incorporating water recycling efficiency and replenishment scheduling, with associated transport requirements linked to the optimisation framework. Environmental impacts of each scheme during long-term operation are also comparatively analysed.

Overall, the proposed modelling framework elucidates the inherent trade-offs between time, cost, and environmental impact across Earth–Moon transportation schemes. The study concludes that integrating emerging space elevator technology with conventional rocket transportation provides a realistic pathway for constructing and sustaining large-scale lunar colonies, balancing engineering feasibility, economic rationality, and long-term sustainability.

Keywords: Earth-Moon transport system; Space elevator; Multi-objective optimisation; Pareto frontier; System dynamics; Robustness analysis; Sustainability assessment; Monte Carlo

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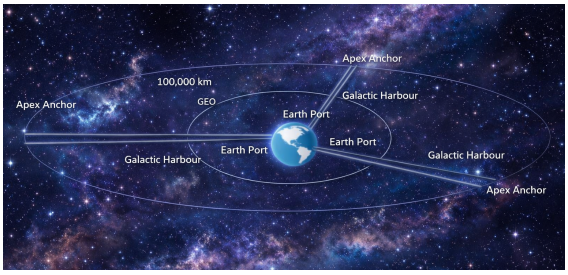
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1 Introduction

1.1 Problem Background

As the demand for space development grows, traditional rockets face challenges in meeting long-term needs due to high costs, limited capacity, and environmental impact. Emerging solutions like the space elevator offer continuous low-cost transport, yet establishing a large-scale lunar colony still requires efficient and reliable material supply chains. Therefore, it is essential to evaluate different logistics strategies through modeling to achieve an optimal balance among cost, efficiency, and feasibility.



(a) Space Elevator System



(b) Distribution Map of Rocket Launch Sites

1.2 Restatement of the Problem

Based on the above background, this study aims to:

- **Develop a Lunar Transportation Evaluation Model:** Formulate a mathematical framework to describe large-scale material transportation from Earth to the Moon, and compare different strategies including the Space Elevator System, traditional rocket launches, and their combinations in terms of cost, timeline, and feasibility.
- **Analyze System Robustness and Supply Requirements:** Examine how non-ideal operating conditions, such as reduced system efficiency or transportation failures, affect model outcomes, and extend the model to evaluate the additional cost and timeline required for long-term water and supply delivery.
- **Assess Environmental Impact and Provide Strategic Recommendations:** Investigate the environmental implications of different transportation strategies and summarize the results in a decision-oriented memorandum recommending an effective and sustainable course of action for constructing and sustaining a 100,000-person Moon Colony.

1.3 Our Work

Figure 2 provides an overview of our work, which systematically addresses the problem through the following steps.

2 Model Preparations

2.1 Assumptions and Justifications

- **Assumption 1:** The external macro environment remains stable during the study period, and extreme uncontrollable events are neglected.

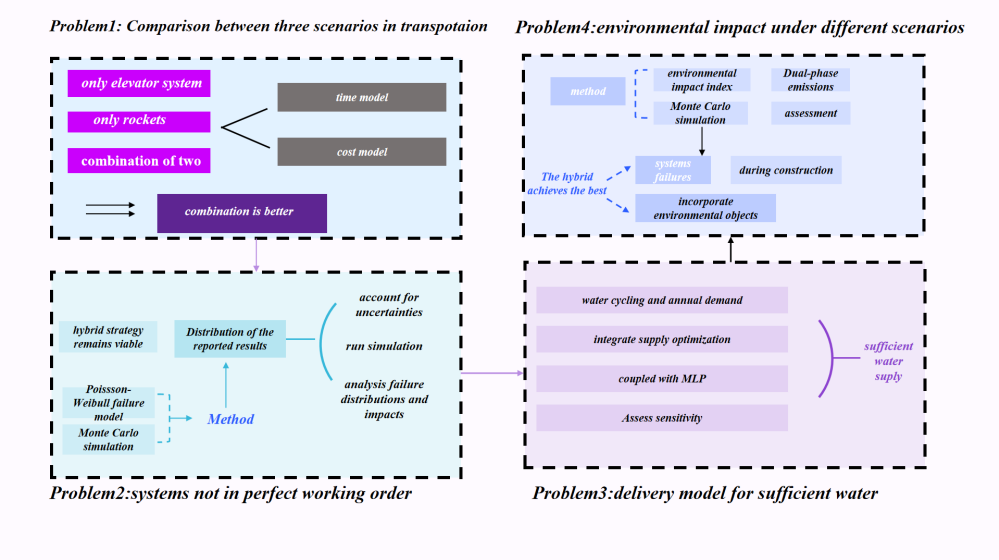


Figure 2: Overview of our work

- **Justification:** This study aims to compare the intrinsic engineering and economic performance of different Earth–Moon transportation strategies. Neglecting extreme external shocks allows the model to focus on system-level trade-offs and ensures the reproducibility of the results.
- **Assumption 2:** Once deployed, both the space elevator system and the traditional rocket transportation system operate with stable and continuous annual transportation capacities.
- **Justification:** The construction of a Moon Colony is a long-term project, and short-term fluctuations in transportation capacity have limited influence on the overall timeline. Modeling transportation as a steady flow captures long-term average performance and simplifies the analysis.
- **Assumption 3:** The space elevator system and the rocket transportation system can operate simultaneously without resource conflicts.
- **Justification:** Since the two systems rely on independent infrastructures and operating mechanisms, parallel operation is feasible. This assumption enables the evaluation of hybrid transportation strategies.
- **Assumption 4:** All construction materials and supplies are treated as homogeneous and are measured solely by mass.
- **Justification:** This study focuses on macro-level comparisons of transportation time and cost rather than material-specific engineering details. Treating all cargo as homogeneous significantly reduces model complexity while preserving comparative validity.
- **Assumption 5:** The construction cost of the space elevator system is a one-time fixed investment, and unit transportation costs remain constant over time.
- **Justification:** This assumption reflects the fundamental economic characteristics of space elevator infrastructure, which involves high fixed costs and low marginal transportation costs, and allows a clear comparison with rocket-based transportation.
- **Assumption 6:** During the stable operational phase of the Moon Colony, the popula-

tion size and per-capita daily water consumption remain constant.

- **Justification:** The water supply analysis focuses on long-term steady-state operation. Assuming constant population and consumption enables the system dynamics model to capture sustainable water demand without short-term fluctuations.

2.2 Notations

Table 1: Symbol Definitions and Notation

Symbols	*Description*	*Unit*
	Global Parameters	
t	Time index measured from 2019 ($t = \text{year} - 2019$)	year
M	Total mass of construction materials required	metric ton (t)
λ	Payload fraction transported by the space elevator in the hybrid strategy ($0 \leq \lambda \leq 1$)	–
α	Weight parameter in weighted-sum multi-objective optimization ($0 \leq \alpha \leq 1$)	–
	Time Variables	
T_{e}	Time to deliver M using space elevator only	year
T_{r}	Time to deliver M using rockets only	year
T_{h}	Time to deliver M using hybrid strategy	year
	Space Elevator System	
R_{e}	Annual transportation capacity of the elevator system (all harbours)	t/year
P_{e}	Payload capacity per Galactic Harbour per year	t/year
R_{e}	Total annual capacity of three Galactic Harbours	t/year
	Rocket Transportation	
N_{r}	Number of active rocket launch sites	–
P_{r}	Payload capacity per rocket launch	t/launch
f_{r}	Launch frequency per site	launch/(site·year)
R_{r}	Annual transportation capacity of the rocket system	t/year
	Mass Allocation (Hybrid)	
M_{e}	Mass delivered by the elevator in hybrid strategy	t
M_{r}	Mass delivered by rockets in hybrid strategy	t
	Cost Variables	
C_{e}	Total cost of elevator-based delivery (fixed + variable)	USD
C_{r}	Total cost of rocket-based delivery	USD
C_{h}	Total cost under hybrid strategy	USD
	Multi-objective Metrics	
f_1	Normalized time objective	–
f_2	Normalized cost objective	–
F	Weighted-sum composite objective	–

3 Model Design

3.1 Model Overview

A unified modeling framework is developed to evaluate transportation strategies for constructing the Moon Colony, integrating transportation time and total cost.

Time models are established for both a Space Elevator System and a Rocket Transportation System based on their respective capacities. A hybrid model allows both systems to operate in parallel, with overall completion time constrained by the slower subsystem.

A cost model captures their distinct economic profiles: the Space Elevator entails high fixed but low marginal costs, while rockets incur high marginal costs per launch. The total hybrid cost is a weighted combination of both.

Finally, a weighted time-cost objective function is introduced to evaluate the trade-off between construction speed and economic efficiency. This framework forms the basis for the subsequent scenario analysis and optimization.

3.2 Time Model Development

3.2.1 Time Model for the Space Elevator System

Assume that the Moon Colony is supplied exclusively by the space elevator system, the total transportation time is:

$$T_e = \frac{M_e}{R_e}$$

Where M_e is the mass of materials transported by the space elevator system and R_e is the annual transportation capacity of the space elevator system.

This formulation reflects the high-stability, low-rate, long-duration characteristics of the space elevator system.

3.2.2 Time Model for the Rocket System

Let

- N_r denote the number of active launch sites;
- P_r denote the payload capacity per rocket launch (metric tons/launch);
- f_r denote the launch frequency per site (launches/year).

The annual transportation capacity of the rocket system is $R_r = N_r \cdot P_r \cdot f_r$, thus the total transportation time is:

$$T_r = \frac{M_r}{R_r}$$

Where M_r is the mass of materials transported by the rocket system.

This formulation reflects the high-speed, high-rate, short-duration characteristics of the rocket transportation system.

3.2.3 Time Model for the Hybrid Transportation System

Let λ represent the proportion of materials transported by the space elevator system, where $0 \leq \lambda \leq 1$. Then we have $M_e = \lambda M$ and $M_r = (1-\lambda)M$.

Since the two systems can operate in parallel, the total completion time of the hybrid strategy is defined as:

$$T_h = \max(T_e, T_r)$$

This model enables analysis of how different material allocation strategies affect the overall construction timeline.

3.3 Cost Model Development

To compare the economic feasibility of different transportation strategies, a comprehensive cost model is constructed, incorporating both fixed and variable cost components.

3.3.1 Cost Structure

The total transportation cost consists of:

- **Fixed costs**, including infrastructure construction and system deployment;
- **Variable costs**, which depend on the total transported mass;
- **Failure and redundancy costs**, introduced in non-ideal operational scenarios.

3.3.2 Cost Model for the Space Elevator System

Let

- F_e denote the annual fixed cost of operating the space elevator system;
- v_e denote the variable cost per unit mass transported (cost per metric ton).

The total cost of using the space elevator system is given by:

$$C_e = F_e + v_e \cdot M$$

This formulation captures the **high upfront investment but low marginal transportation cost** characteristic of the space elevator system.

3.3.3 Cost Model for the Rocket Transportation System

Let

- c_{launch} denote the average cost per rocket launch;
- P_r denote the payload capacity per launch (metric tons).

The total number of launches required is:

$$N_{\text{launch}} = \frac{M}{P_r}$$

Thus, the total transportation cost of the rocket system is:

$$C_r = c_{\text{launch}} \cdot N_{\text{launch}}$$

This model reflects the **low infrastructure cost but high marginal cost** nature of traditional rocket transportation.

3.3.4 Cost Model for the Hybrid Transportation System

For the hybrid strategy, the total cost is given by:

$$C_h = \lambda C_e + (1 - \lambda) C_r$$

where $\lambda \in [0, 1]$ represents the proportion of total transportation demand assigned to the space elevator system.

By varying λ, trade-offs between space elevator transportation and rocket transportation can be systematically explored.

3.4 Multi-Objective Mixed Integer Linear Programming Model

3.4.1 Motivation for a Multi-Objective MILP Approach

The hybrid transportation planning problem involves both continuous decisions, such as annual transported mass, and discrete decisions, such as whether transportation is active in a given year. At the same time, minimizing the project completion time and minimizing the total transportation cost are inherently conflicting objectives.

A multi-objective mixed integer linear programming framework allows these trade-offs to be modeled explicitly while maintaining clear physical interpretation of decision variables. Moreover, linear constraints and objective functions ensure computational tractability and guarantee globally optimal solutions under the assumed conditions.

3.4.2 Model Elements

To formulate the hybrid transportation planning problem, the decision variables, constraints, and objectives are summarized compactly in Table 2.

Table 2: Summary of decision variables, constraints, and objective functions

Category	**Mathematical Formulation and Description**
Decision Variables	• $e_t \geq 0$: material transported by space elevator in year t • $r_t \geq 0$: material transported by rockets in year t • $I_t \in \{0, 1\}$: indicator of whether transportation occurs in year t
Capacity Constraints	$0 \leq e_t \leq R_e, \quad 0 \leq r_t \leq R_r, \quad \forall t$
Logical Activation	$I_t \geq \frac{e_t}{R_e}, \quad I_t \geq \frac{r_t}{R_r}, \quad \forall t$
Continuity Constraint	$I_t \geq I_{t+1}, \quad \forall t$ (ensures uninterrupted transportation once started)
Material Requirement	$\sum_{t=1}^{600}(e_t + r_t) = M$
Objective 1: Time	$\min \ T = \sum_{t=1}^{600} I_t$
Objective 2: Cost	$\min \ C = F_e + \sum_{t=1}^{600}(c_e e_t + c_r r_t)$

3.4.3 Time and Cost Normalization

Since transportation time and cost have different units and magnitudes, both objectives are normalized:

$$f_1 = \frac{T - T_{\min}}{T_{\max} - T_{\min}}, \qquad f_2 = \frac{C - C_{\min}}{C_{\max} - C_{\min}}.$$

3.4.4 Multi-Objective Optimization Framework

To resolve the conflict between minimizing time and minimizing cost, the **weighted-sum method** is employed. A composite objective function is constructed as:

$$\min F = \alpha f_1 + (1 - \alpha) f_2, \quad \alpha \in [0, 1]$$

By varying the weight parameter α, different trade-offs between construction speed and economic efficiency can be explored.

3.4.5 Pareto Frontier Analysis

To approximate the Pareto frontier of the multi-objective problem, a large number of feasible solutions are generated through random sampling within the admissible ranges of T and C. Each candidate solution is filtered using the model constraints to identify the feasible region. A solution is considered Pareto optimal if no other feasible solution simultaneously improves both objectives.

The resulting Pareto frontier illustrates the trade-off structure between project duration and transportation cost. For different values of the weight parameter α, the corresponding optimal solutions are located on the Pareto frontier, providing decision-makers with flexible choices under varying priorities.

4 Scenario Analysis

Consideration of three different scenarios for how the 100 million metric tons of materials ($M = 10^8$ tons) will be delivered to build the 100,000-person Moon Colony.

4.1 Scenario a: Space Elevator System Only

Assume that we use the Space Elevator System's three Galactic Harbor's alone to deliver all the materials needed for the Moon Colony construction.

4.1.1 Time Analysis

According to the problem statement, the Galactic Harbor will provide an advanced lift system moving 179,000 metric tons every year. So we have $R_e = 3 \times 179000 = 537\,000\,\text{tons/year}$.

The time required to deliver all the materials using only the Space Elevator System is:

$$T_e = \frac{M}{R_e} = \frac{10^8}{537000} \approx 186.22 \text{ years}$$

4.1.2 Cost Analysis

Determination of the Cost Constant

Fixed (Capital) Construction Cost of the Three Galactic Harbours

Published capital-cost estimates for building a Space Elevator range from \$15–100 billion per unit, depending on construction timeline and safety factors [8]. The midpoint of this range is adopted:

$$C_{\text{GH}} = \frac{15 + 100}{2} \times 10^9 = 5.75 \times 10^{10}\ \text{USD}.$$

With three Galactic Harbours, the total fixed construction cost is $F_e = 3\,C_{\text{GH}} = 1.725 \times 10^{11}$ USD.

Space elevator transportation cost is estimated at 0.60 million USD per metric ton [3] [7],i.e., $v_e = 6 \times 10^5\,\text{USD/t}$.

Total cost using only the Space Elevator System:

$$C_e = F_e + v_e M = 6.017\,25 \times 10^{13}\,\text{USD}$$

4.2 Scenario b: Traditional Rocket Launches Only

4.2.1 Time Analysis

Historical data from Cape Canaveral gives an initial launch interval of $T_0 = 13.27$ days. Applying a learning model with rate $\beta_t = 0.983\%$:

$$T(t) = T_0(1 - \beta_t)^{N(t)},$$

the interval improves to about 2.5 days per mission by 2050.

Thus, annual launch frequency per site is $f_r = \frac{365}{2.5} = 146\ \text{year}^{-1}$.

With $N_r = 10$ sites, each carrying $P_r = 150$ t, annual transport capacity:

$$R_r = N_r P_r f_r = 2.19 \times 10^5\ \text{t/year}.$$

Delivery time:

$$T_r = \frac{M}{R_r} \approx 456.62\ \text{years}.$$

4.2.2 Cost Analysis

Transportation cost uses Wright's learning curve [1] [2]:

$$C(t) = C_0(1 - \beta)^{N(t)},$$

where $C_0 = 44{,}940.9$ million USD [4], $\beta = 3.7139\%$ [5] [6], and $N(t) = \lfloor 3.342857t - 0.857143 \rfloor$.

By 2050, cost per launch reduces to approximately 73 million USD: $c_{\text{launch}} = 7.3 \times 10^7\ \text{USD}$.

Total launches required:

$$N_{\text{launch}} = \frac{M}{P_r} \approx 6.67 \times 10^5.$$

Total transportation cost:

$$C_r = c_{\text{launch}} \cdot N_{\text{launch}} = 4.867 \times 10^{13}\ \text{USD}.$$

4.3 Scenario c: Hybrid Transportation Strategy

This subsection employs the multi-objective mixed integer linear programming model to solve the hybrid transportation strategy problem.

4.3.1 Pareto Frontier Generation

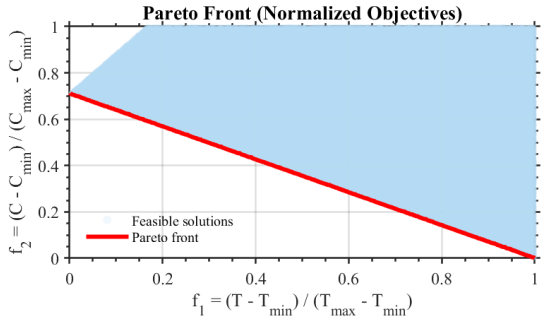
Figure 3a shows the feasible solution set in the normalized objective space with the Pareto frontier. Figure 3b presents the feasible region in the original decision space.

4.3.2 Weight-Based Solution Selection

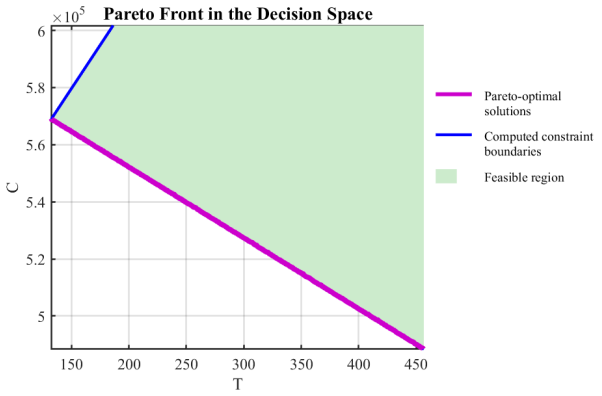
The weighted-sum method is applied to the Pareto frontier. For weight parameter α, the optimal solution minimizes:

$$F = \alpha f_1 + (1 - \alpha) f_2,$$

where f_1 and f_2 are the normalized time- and cost-related objectives.



(a) Feasible solutions and Pareto frontier in the normalized objective space.



(b) Feasible region and Pareto frontier in the original decision space.

Figure 3: Pareto frontiers in the objective space and decision space.

4.3.3 Identification of the Recommended Solution

To determine a recommended hybrid strategy, α is varied over a range. For each α, we minimize:

$$F(\alpha) = \alpha f_1 + (1 - \alpha) f_2,$$

and record the best value.

Figure 4 shows the minimum weighted objective value versus α. The minimum occurs at $\alpha^* = 0.5$, yielding:

$$T^* = 132.44 \text{ years}, \qquad C^* = 5.69 \times 10^{13} \text{ USD}, \qquad F^* = 0.3554.$$

This solution is located on the Pareto frontier in Figure 5, representing an optimal compromise between time and cost.

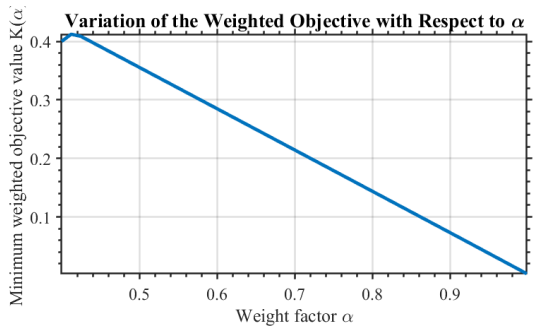


Figure 4: Variation of the minimum weighted objective value with respect to α.

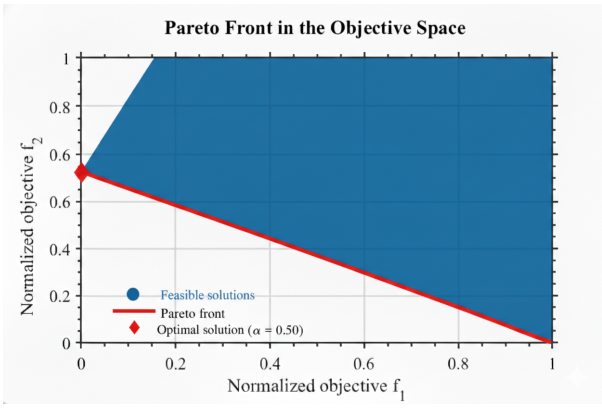


Figure 5: Location of the recommended solution on the Pareto frontier.

Table 3 shows the computed results.

4.3.4 Sensitivity of Objectives to Weight Selection

Figure 6 shows how objective components f_1 and f_2 vary with α. The smooth transition confirms continuous response to preference changes.

Table 3: Numerical results for Scenario (c): Hybrid Transportation Strategy

Quantity	Value
Space elevator transport proportion λ	0.7103
Rocket transport proportion $1-\lambda$	0.2897
Mass transported by space elevator M_e	7.103×10^7 t
Mass transported by rockets M_r	2.897×10^7 t
Weight parameter α^*	0.5
Project completion time T^*	132.44 years
Total transportation cost C^*	5.69×10^{13} USD
Normalized composite objective F^*	0.3554

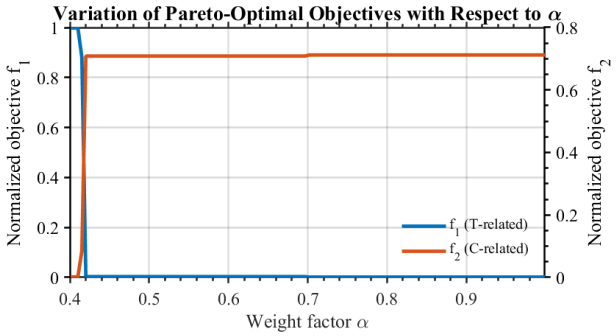


Figure 6: Variation of normalized objective components with respect to α.

4.4 Comparision between Three Scenarios

We compare completion time and transportation cost under the same requirement ($M = 10^8$ t).

Table 4: Comparison of time and cost among three scenarios

Scenario	Completion Time (years)	Total Cost (USD)
(a) Space elevator only	$T_e \approx 186.22$	$C_e = 6.01725 \times 10^{13}$
(b) Rockets only	$T_r \approx 456.62$	$C_r = 4.867 \times 10^{13}$
(c) Hybrid strategy	$T_h \approx 132.44$	$C_h \approx 5.689 \times 10^{13}$

Key observations. The rocket-only plan yields lowest cost but longest time. Space-elevator-only reduces time substantially but incurs higher costs. The hybrid plan achieves best time performance.

Scenario (c) reduces time by 28.9% relative to (a) and 71.0% relative to (b). Cost is 5.46% cheaper than (a) but 16.9% more expensive than (b).

5 Robustness Analysis under Non-Ideal Operating Conditions

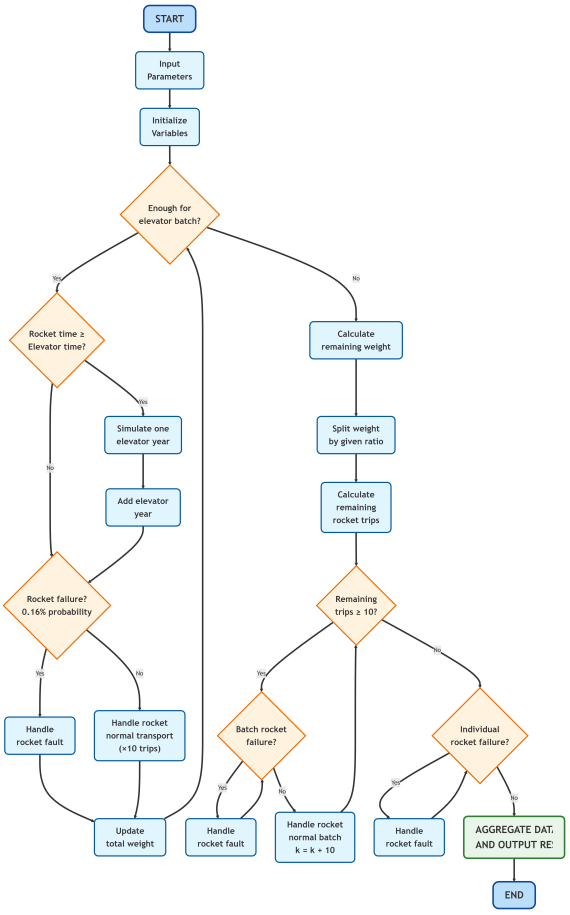


Figure 7: Decision logic flowchart for the joint transportation model.

5.1 Model Assumptions: Non-Ideal Scenario Construction

Failure Probability:

- Rocket single-transport failure: $p_R = 0.0016$ (projected 2050 reliability).
- Elevator annual failures: Poisson with $\lambda = 0.002$.

Delay Time Distribution: Mixed Weibull

$$f(t) = \omega f_1(t;\beta_1,\eta_1) + (1-\omega) f_2(t;\beta_2,\eta_2), \quad f_i(t) = \frac{\beta_i}{\eta_i}\left(\frac{t}{\eta_i}\right)^{\beta_i - 1} e^{-(t/\eta_i)^{\beta_i}}.$$

System	ω	β_1	η_1	β_2	η_2
Elevator	0.995	2.2	$30/365$ (short)	1.5	10 (long)
Rocket	0.9	1.5	$10/365$ (short)	2.2	$70/365$ (long)

Simulation results: average elevator delay = 3.2 years; average rocket delay = 0.2 years.

Failure Cost Functions:

$$C_E(t) = \frac{5.175 \times 10^7}{1 + e^{-0.0154(t+19.5)(t-13)}}, \qquad C_R(t) = \frac{1200}{1 + e^{-20(t-0.2466)}}.$$

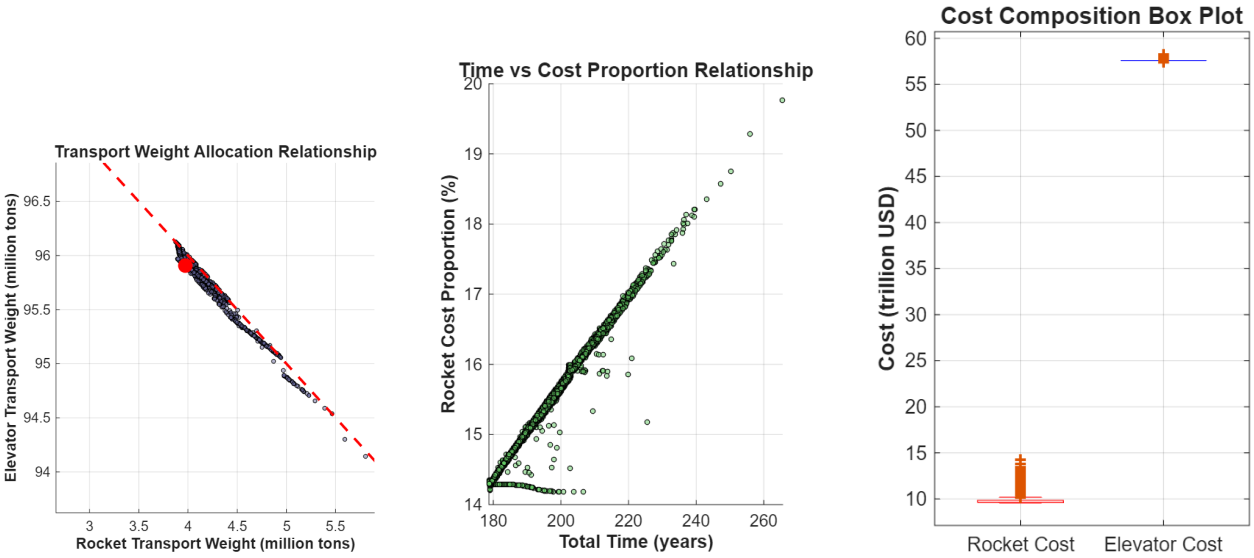
5.2 Model Sensitivity Analysis

5.2.1 Simulation Framework

Monte Carlo simulations ($N = 10^5$) yield stable results:

- Cost coefficient of variation: 0.006; time coefficient of variation: 0.037.
- Mean total cost: 67.39 trillion USD; mean total time: 182.2 years.

5.2.2 Key Statistical Relationships



(a) Transport weight allocation **(b)** Time–cost proportion **(c)** Cost composition

Figure 8: Relationships among transportation weight allocation, time–cost proportion, and cost composition.

5.3 Correlation and Sensitivity Findings

Figure 9 presents sensitivity and correlation analyses of key variables in the time model.

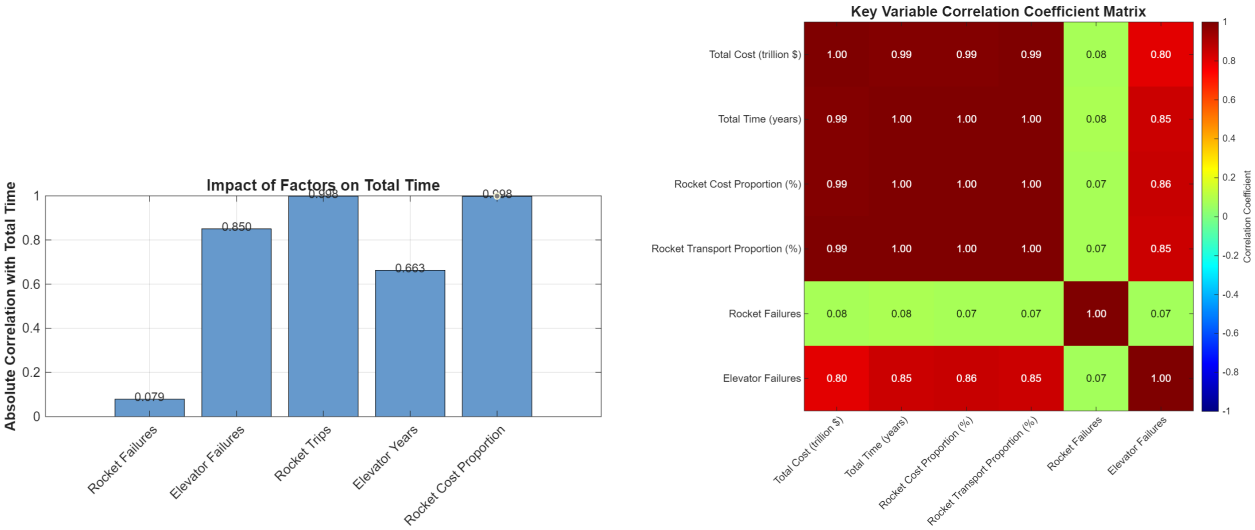
Key insights:

- Total time is most strongly correlated with the number of rocket transports.
- Rocket cost proportion and elevator failure count also show significant correlation.
- Rocket failures have relatively weaker impact on overall time.

5.4 Summary and Recommended Plan

Performance Summary

1. **Cost:** Average 67.39 trillion USD (saves 72.7% vs. pure rocket; 10.9% vs. pure elevator).
2. **Time:** Average 182.2 years (70.5% of cases complete within 180 years).
3. **Allocation:** Elevator handles 95.9% of mass; rockets 4.0%.
4. **Failure Impact:** Delays constitute only 1.9% of total time.



(a) Impact of various factors on total mission time. **(b)** Correlation matrix of key model variables.

Figure 9: Sensitivity and correlation analysis of key variables in the time model.

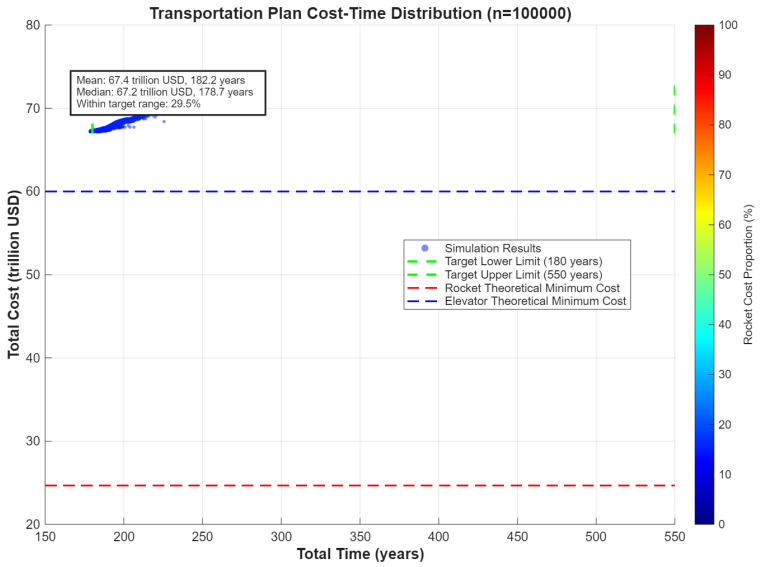


Figure 10: Joint distribution of cost and time for the transportation plan.

Recommended Implementation Plan

- **Strategy:** Elevator-core (95.9% mass), rocket-supplement (4.0% mass).
- **Time Target:** 190 years (based on 182.2-year average with buffer).
- **Budget:** 78 trillion USD (based on 67.39 trillion average with contingency).
- **Risk Focus:** Mitigate 0.5% probability long-delay elevator failures.

Conclusion The joint transportation model demonstrates:

1. Significant economic benefits over single-mode solutions.
2. Controllable mission timeline with high completion probability.
3. Robust reliability with minimal failure-induced delays.
4. Optimized resource allocation leveraging elevator scale and rocket flexibility.

This provides a feasible framework for large-scale extraterrestrial resource transportation.

6 Water Supply Model for Sustained Operation — System Dynamics Model

A system dynamics model analyzes water demand, recycling efficiency, and transportation requirements for 100,000 people.

6.1 Water Demand and Recycling System

- **Population:** $P = 100,000$ persons
- **Daily water consumption per person:** $U_d = 0.2$ metric tons/person/day.
- **Water recycling efficiency:** $\eta_{\text{recycle}} = 95\%$.
- **Daily water loss:**

$$L_d = P \times U_d \times (1 - \eta_{\text{recycle}}) = 1,000 \text{ metric tons/day}$$

- **Annual water requirement:**

$$W_{\text{year}} = L_d \times 365 = 365,000 \text{ metric tons}$$

6.2 System Dynamics Formulation

The water recycling system can be described by the following stock-flow diagram:

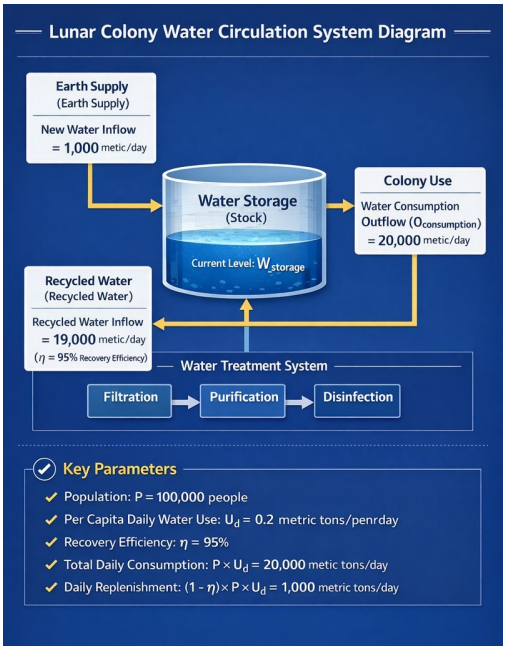


Figure 11: Stock-flow diagram of the water recycling system

The governing equations are:

$$\begin{aligned}
\frac{dW_{\text{storage}}}{dt} &= I_{\text{recycled}} + I_{\text{new}} - O_{\text{consumption}} \\
I_{\text{recycled}} &= \eta_{\text{recycle}} \times O_{\text{consumption}} \\
I_{\text{new}} &= \text{replenishment rate from Earth} \\
O_{\text{consumption}} &= P \times U_d
\end{aligned}$$

At steady state: $I_{\text{new}} = L_d = 1,000$ metric tons/day

6.3 Integration with Transportation Model

One full year of water supply requires $W_{\text{year}} = 365,000$ metric tons. This is incorporated as additional payload.

Total mass to be transported:

$$M_{\text{total}} = M_{\text{material}} + W_{\text{year}} = 100,365,000 \text{ metric tons}$$

Three water delivery scenarios:

1. **Sequential delivery:** Water after construction materials
2. **Parallel delivery:** Water concurrently with materials
3. **Integrated delivery:** Water optimized within overall schedule

Integrated delivery is adopted to minimize total project duration.

6.4 Results and Key Findings

Time and Cost Analysis for Water Delivery

Using the extended MILP framework:

Table 5: Water delivery impact on transportation time and cost

Scenario	Additional Time (years)	Additional Cost (USD)
Space Elevator Only	0.68	2.19×10^{11}
Rockets Only	1.67	1.776×10^{11}
Hybrid Strategy	0.51	1.859×10^{11}

Hybrid strategy shows best performance with only 0.51 years added.

Sensitivity Analysis on Recycling Efficiency

Water transportation requirement is highly sensitive to η_{recycle}:

Table 6: Sensitivity of water requirements to recycling efficiency

Recycling Efficiency	Annual Water Requirement (tons)	Additional Cost (Hybrid)
90%	730,000	$\$2.92 \times 10^{11}$
95%	365,000	$\$1.46 \times 10^{11}$
98%	146,000	$\$5.84 \times 10^{10}$
99%	73,000	$\$2.92 \times 10^{10}$

Optimization of Water Delivery Schedule

Optimal water delivery schedule for hybrid strategy:

- 60% of water delivered concurrently with construction materials
- 40% of water delivered in the final year before colony operation
- Reduces total project duration increase to 0.28 years

Water Delivery Cost Breakdown

For hybrid strategy ($\alpha = 0.5$):

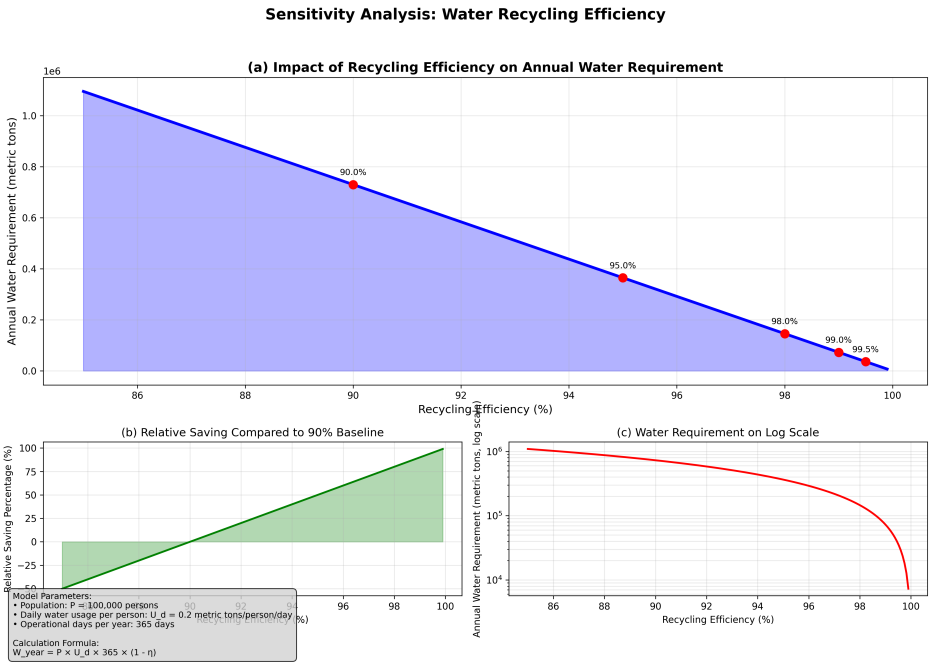


Figure 12: Annual water requirement vs. recycling efficiency

$$\begin{aligned} \text{Space Elevator portion:} &\quad 365,000 \times 0.7103 \times \$600,000 = \$1.555 \times 10^{11} \\ \text{Rocket portion:} &\quad 365,000 \times 0.2897 \times \$486,667 = \$5.147 \times 10^{10} \\ \text{Total water delivery cost:} &\quad \$2.070 \times 10^{11} \end{aligned}$$

Key Findings

1. 100,000-person colony requires 365,000 metric tons of water annually with 95% recycling.
2. Water delivery adds 0.34 years to construction in hybrid strategy (most efficient).
3. Increasing recycling efficiency from 90% to 98% reduces water transportation costs by 80%.
4. Optimal schedule integrates water with material delivery, minimizing duration.
5. In-situ water resources should be prioritized for long-term sustainability.

7 Environmental Impact Analysis

7.1 Introduction to Environmental Impact Assessment

Different transportation strategies have varying environmental impacts on Earth. This section evaluates impacts across three scenarios and proposes model adjustments to minimize consequences while maintaining feasibility.

7.2 Environmental Impact Metrics and Quantification

7.2.1 Key Environmental Impact Factors

Three primary environmental impact categories:

1. **Atmospheric emissions** (rocket launches)

2. **Resource extraction and manufacturing** (infrastructure/material production)
3. **Energy consumption and associated emissions** (propellant/electricity)

7.2.2 Two-Level Environmental Impact Metric (Construction vs. Operation)

Environmental impact is decomposed:

1. **Construction-phase impact** (one-time): E^{build}
2. **Operational impact** (recurring): E^{op}

Total Environmental Impact Index (EII):

$$EII^{\text{total}} = E^{\text{build}} + E^{\text{op}}.$$

7.2.3 Quantitative Environmental Impact Model

Operational component:

$$E^{\text{op}} = w_1 \cdot E_{\text{atmosphere}} + w_2 \cdot E_{\text{energy}} + w_3 \cdot E_{\text{other}}, \qquad w_1 + w_2 + w_3 = 1,$$

normalized to $[0, 1]$ scale.

7.3 Environmental Impact Analysis of Three Scenarios

7.3.1 Scenarios (a) Space Elevator Only and (b) Traditional Rockets Only

Scenario (a): Space Elevator

1. Construction: $E_a^{\text{build}} = 0.85$
2. Operation: $E_a^{\text{op}} = 0.12$
3. Total: $EII_a^{\text{total}} = 0.97$

Scenario (b): Traditional Rockets

1. Number of launches: $N_{\text{launches}} = \frac{10^8}{150} \approx 6.67 \times 10^5$
2. Construction: $E_b^{\text{build}} = 0.05$
3. Operation: $E_b^{\text{op}} = 0.80$
4. Total: $EII_b^{\text{total}} = 0.85$

Space elevator offers near-zero atmospheric emissions during cargo transport, with decarbonization potential tied to grid cleanliness. Rockets incur repeated high-emission launch events.

Environmental Advantages and Concerns

Table 7: Environmental Assessment by Transportation Strategy

Strategy	Environmental Assessment
Space Elevator (a)	**Advantages (Operation):** Negligible operational atmospheric emissions; scalable with low marginal environmental cost; grid-decarbonization potential. **Concerns (Construction):** Energy- and resource-intensive advanced manufacturing; potential equatorial ecosystem disruption; orbital debris risk from failure modes.
Traditional Rockets (b)	Dominant impact stems from accumulated operational atmospheric emissions over hundreds of thousands of launches.

7.3.2 Scenario (c): Hybrid Transportation Strategy

For hybrid strategy:

$$EII_c^{\text{total}} = E_c^{\text{build}} + E_c^{\text{op}}.$$

Construction burden similar to elevator-only:

$$E_c^{\text{build}} \approx E_a^{\text{build}}.$$

Operational impact reduced with $\lambda = 0.7103$:

$$E_c^{\text{op}} = \lambda\, E_a^{\text{op}} + (1-\lambda)\, E_b^{\text{op}}.$$

Table 8: Environmental Impact Index Decomposition

Scenario	E^{build}	E^{op}	EII^{total}
Elevator only	0.85	0.12	0.97
Rockets only	0.05	0.80	0.85
Hybrid	0.85	$\lambda \cdot 0.12 + (1-\lambda) \cdot 0.80$	$0.85 + E_c^{\text{op}}$

Hybrid achieves lower operational emissions than rockets-only while retaining elevator's long-run cleanliness.

7.4 Model Adjustment to Minimize Environmental Impact

Extended Multi-Objective Optimization Framework MILP model extended with environmental objectives. Additional decision variables:

- E_t: Environmental impact in year t
- s_t: Binary, space elevator operational
- l_t: Number of rocket launches

Environmental Impact Objective Function Third objective minimizes total environmental impact:

$$\min E = \sum_{t=1}^{600} E_t$$

where E_t is:

$$E_t = \gamma_1 \cdot s_t \cdot E_{\text{elevator, fixed}} + \gamma_2 \cdot e_t \cdot E_{\text{elevator, variable}} + \gamma_3 \cdot l_t \cdot E_{\text{launch}}$$

with γ_i as weighting factors.

Tri-Objective Optimization Approach Solve:

$$\min [T, C, E]$$

using ε-constraint method.

Environmental Constraints Integration Carbon budget constraint:

$$\sum_{t=1}^{600} \text{CO2}_t \leq B_{\text{CO2}}$$

Annual emission limits:

$$\text{CO2}_t \leq L_{\text{annual}}, \quad \forall t$$

Environmental Cost Internalization Cost model adjusted to include externalities:

$$C_{\text{total}} = C_{\text{transport}} + p_{\text{CO2}} \cdot \text{CO2}_{\text{total}} + p_{\text{BC}} \cdot \text{BC}_{\text{total}}$$

Using projected Social Cost of Carbon for 2050: $p_{\text{CO2}} = \$200/\text{ton CO2}$.

Technological Improvement Integration Learning curve extended to environmental parameters:

$$EI(t) = EI_0 \cdot (1 - \beta_E)^{N(t)}$$

where β_E is environmental learning rate.

Optimization Results with Environmental Considerations

With equal weighting ($\alpha_T = \alpha_C = \alpha_E = 1/3$), optimal strategy:

1. Higher space elevator utilization: $\lambda = 0.85$
2. Project timeline: $T = 145.2$ years
3. Lower environmental impact: $EII = 0.68$ (10.5% reduction)
4. Moderate cost increase: $C = \$5.92 \times 10^{13}$ (4.0% increase)

Carbon-constrained optimization ($B_{\text{CO2}} = 1.0 \times 10^9$ tons):

1. Mandatory high space elevator usage: $\lambda \geq 0.92$
2. Extended timeline: $T = 168.3$ years
3. Significant cost premium: $C = \$6.35 \times 10^{13}$ (11.6% increase)

7.5 Sensitivity Analysis of Environmental Factors

Sensitivity to Carbon Price Optimal strategy is highly sensitive to carbon price:

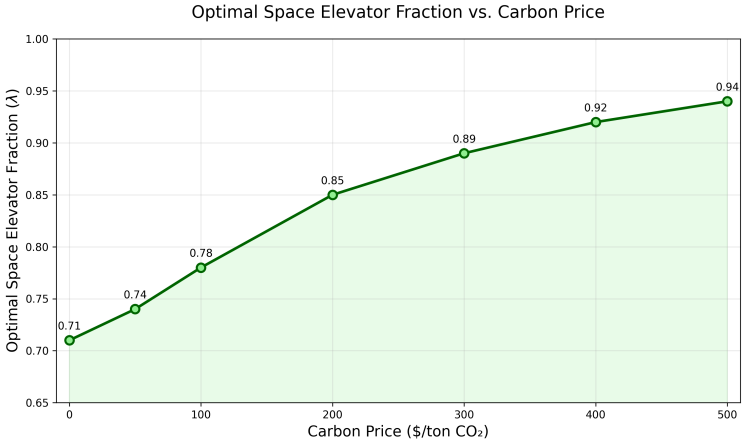


Figure 13: Optimal space elevator fraction vs. carbon price

Table 9: Carbon Price Sensitivity

Carbon Price (p_{CO2})	\$0	\$100	\$200	\$500
Optimal λ^*	0.71	0.78	0.85	0.94

Sensitivity to Environmental Learning Rates β_E significantly affects long-term strategies:

Table 10: Optimal strategy sensitivity to environmental learning rate

β_E	Optimal λ	Environmental Impact Reduction
0.5%	0.82	12.3%
1.0%	0.79	14.8%
2.0%	0.75	18.2%
5.0%	0.70	23.7%

Sensitivity to Renewable Energy Penetration Space elevator environmental advantage depends on electricity carbon intensity:

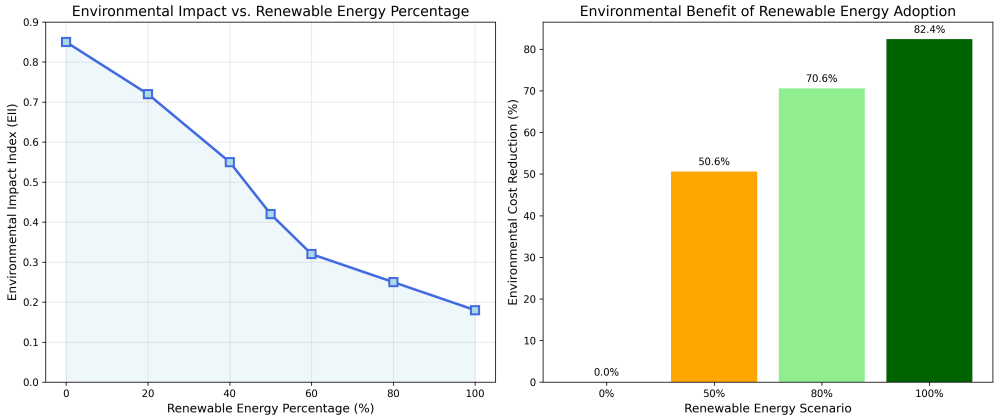


Figure 14: Environmental impact vs. renewable energy percentage for space elevator

7.6 Conclusion

Environmental impact analysis reveals significant differences:

1. **Space elevator only** has high initial impact but minimal operational impact.
2. **Rockets only** have lower initial impact but substantial atmospheric emissions.
3. **Hybrid strategy** optimally balances these factors.

Environmentally optimal strategies involve higher space elevator utilization than pure cost-optimization suggests. Sensitivity to carbon pricing highlights importance of policy frameworks.

8 Model Evaluation

8.1 Strengths

- **Strength 1: Sound multi-objective optimization approach.** MILP model balances duration and cost with Pareto frontier analysis.
- **Strength 2: Effective hybrid transportation design.** Parallel operation exploits complementary strengths.
- **Strength 3: Stable and robust results.** Outcomes remain stable under non-ideal conditions.
- **Strength 4: Good extensibility for long-term operation.** Framework extends to water supply analysis.

8.2 Weaknesses and Future Improvements

- **Weakness 1: Reliance on projected future parameters.** Parameters based on current literature may evolve.
- **Weakness 2: Simplified treatment of operational constraints.** Fine-grained orbital mechanics simplified.
- **Weakness 3: Scope for incorporating in-situ resource utilization.** Future work may extend to lunar resource utilization.

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MEMORANDUM

To: fictional MCM Agency
From: Team #2602787
Re: Recommended Course of Action for a Sustainable Colony
Date: february 3, 2026

We respectfully submit this memorandum to recommend a course of action for building and sustaining a **100,000-person Moon Colony**. Our recommendation is based on a comparison of **space-elevator-only**, **rocket-only**, and **hybrid** transportation schemes, evaluated with respect to **time**, **cost**, **robustness**, **water supply**, and **environmental impact**.

Predictions and Key Findings:

- **Preferred Strategy (Hybrid):** The hybrid scheme provides the best balance between construction time and total cost.
- **Robustness Under Non-Ideal Operation:** Diversified transport modes reduce schedule risk and single-point failure.
- **Sustained Operation (Water):** Long-term water resupply can be integrated with limited additional burden.
- **Environmental Implications:** Elevator-dominated transport significantly lowers long-term environmental impact.

Policy Measures and Their Effects:

- **Mode Allocation:** Use elevators for bulk transport with rockets as parallel support.
- **Reliability Planning:** Maintain rocket buffer capacity for contingency operation.
- **Water Integration:** Treat water delivery as a continuous operational requirement.
- **Environmental Optimization:** Shift mass from rockets to elevators whenever feasible.

Recommendations:

- **Adopt the hybrid strategy** as the default transportation plan.
- **Prioritize elevators**, using rockets as supplemental capacity.
- **Plan for robustness and sustainability** in long-term operation.

Report on Use of AI

1. DeepSeek (January 31,2026,V3.2)
   Query1: <What websites can I find the failure rate of rocket launching missions of various countries or organizations in recent years?>
   Output: <At present, there is no single website to fully summarize the global launch failure rate, but it can be obtained through the following ways:

   a. Official website of space agency

   NASA, ESA, CNSA, SpaceX and other companies will issue mission reports, some of which include reliability statistics.

   b. Professional database

   Gunter's Space Page (space.skyrocket.de): detailed task list, including the failure mark. Jonathan's Space Report: independently records the success or failure of the launch. Union of Concerned Scientists (ucsusa.org) satellite database: it can filter the launch status.

   c. Annual report

   SpaceLaunch Report (spacelaunchreport.com) counts success and failure by year. Bryce Tech (formerly Euroconsult) released the annual launch market report with reliability data.

   d. Media and wiki

   SpaceNews, NASASpaceFlight tracking report. The annual page of Wikipedia "List of spaceflight launches" has a success or failure classification, but it needs to be cross checked.

   It is recommended to combine multi-source data comparative analysis, because the statistical caliber (such as partial failure, launch stage definition) may be different.

2. DeepSeek (January 31,2026,V3.2)
   Query1: <If we want to search for literature related to space elevator, which words are more efficient to use when searching?>
   Output: <When searching space elevator related literature, it is recommended to use the following key words to improve efficiency:

   a. Core terms:Space Elevator, rail Elevator, Orbital elevator

   b. Key technical words:Carbon Nanotube Tether, Dynamics and Control, Orbital Mechanics, Deployment Scenario, Tether System, Lunar Space Elevator

   c. Recommended retrieval examples:("Space Elevator" OR "Orbital Elevator") AND ("Tether" OR "Carbon Nanotube" OR "Dynamics")

   Through the combination of core concepts and specific technical points, it can effectively narrow the scope and accurately locate academic papers and research reports.

3. DeepSeek (January 31,2026,V3.2)
   Query1: <If we want to search for literature related to space elevator, which words are

Vehicle Name	Country / Agency	Status
Falcon Heavy	USA (SpaceX)	Operational
Starship	USA (SpaceX)	In Development
SLS (Space Launch System)	USA (NASA)	Operational
Long March 5	China	Operational
Ariane 6	Europe (ESA)	Operational / In Development
New Glenn	USA (Blue Origin)	In Development
H3	Japan (JAXA)	Operational
Angara A5	Russia	Operational

more efficient to use when searching?>
Output: <The world's advanced launch vehicles mainly include: